Information Processing in Agriculture 12 (2025) 54–67

Contents lists available at ScienceDirect

Information Processing in Agriculture

journal homepage: www.keaipublishing.com/en/journals/information-processing-in-agriculture/

Innovative deep learning approach for cross-crop plant disease detection: A generalized method for identifying unhealthy leaves

Imane Bouacida [a,*], Brahim Farou [a], Lynda Djakhdjakha [a], Hamid Seridi [a], Muhammet Kurulay [b]

[a] LabSTIC Laboratory, Computer Science Department, 8 Mai 1945 Guelma University, Guelma, 24000, Algeria
[b] Istanbul Technical University, Department of Mathematical Engineering, 34469, Maslak, Istanbul, Türkiye

A R T I C L E I N F O

Keywords:
Deep learning
Agriculture
Plant diseases
Image processing
Disease detection
Leaf splitting

A B S T R A C T

One of the most serious threats to global food security is plant diseases compromising agricultural productivity and threatening the livelihoods of millions. These diseases can decimate crops, disrupt food supply chains, and escalate the risk of food shortages, underscoring the urgency of implementing robust strategies to safeguard the world's food sources. Deep learning methods have revolutionized the field of plant disease detection, offering advanced and accurate solutions for early identification and management. However, a recurring problem in deep learning models is their susceptibility to a lack of robustness and generalization when facing novel crop and disease types that were not included in the training dataset. In this paper, we address this issue by proposing a novel deep learning-based system capable of recognizing diseased and healthy leaves across different crops, even if the system was not trained on them. The key idea is to focus on recognizing the diseased small leaf regions rather than the overall appearance of the diseased leaf, along with determining the disease's prevalence rate on the entire leaf. For efficient classification and to leverage the excellence of the Inception model in disease recognition, we employ a small Inception model architecture, which is suitable for processing small regions without compromising performance. To confirm the effectiveness of our method, we trained and tested it using the widely acclaimed PlantVillage dataset, recognized as the most utilized dataset for its comprehensive and diverse coverage. Our method achieved an accuracy rate of 94.04%. Furthermore, when tested on new datasets, it achieved an accuracy rate of 97.13%. This innovative approach not only enhances the accuracy of plant disease detection but also addresses the critical challenge of model generalization to diverse crops and diseases. In addition, it outperformed the existing methods in its ability to identify any disease across any crop type, showcasing its potential for broad applicability and contribution to global food security initiatives.

1. Introduction

Agriculture is a basic source of food, guaranteeing the continuity of life, and it is also an important component of all economic systems. Improving the agriculture sector is essential to maximizing production and increasing quality. This involves providing the necessary conditions to ensure the healthy growth of plants and crops. Diseases are usually the main causes of plant deterioration. According to estimates by the Food and Agriculture Organization of the United Nations (FAO), these diseases impose an annual cost of approximately $220 billion on the global economy. They result in damage and sometimes the total eradication of crops. Indeed, bacteria, fungi, microscopic animals, and viruses hit the plants, affect their vital function, and cause changes in their natural form [1].

Early detection of diseases allows for neutralizing them and saving plants. The sooner they are discovered, the more losses are avoided and limited, and the crops are protected. The traditional ways used to detect diseases, mainly based on human diagnosis, are insufficient due to a lack of knowledge and experience, and they are time-consuming [2]. Additionally, the frequency of verification and the method used to collect data are inadequate; therefore, the diagnosis must be based on a more reliable method. For this purpose, modern technologies have been introduced as an automatic solution to identify plant diseases [3]. Among these technologies, cutting-edge tools such as robots,

* Corresponding author.
E-mail addresses: bouacida.imane@univ-guelma.dz (I. Bouacida), farou.brahim@univ-guelma.dz (B. Farou), djakhdjakha.lynda@univ-guelma.dz (L. Djakhdjakha), seridi.hamid@univ-guelma.dz (H. Seridi), muhammetkurulay@gmail.com (M. Kurulay).

https://doi.org/10.1016/j.inpa.2024.03.002
Received 3 October 2023; Received in revised form 5 February 2024; Accepted 1 March 2024
Available online 2 March 2024

drones, and sensors have revolutionized field management for farmers [4], while machine learning has created new opportunities for data analysis [5].

The application of machine learning and deep learning techniques in plant disease detection is a rapidly evolving field, demonstrating promising results [6]. However, deep learning-based methods have proved their effectiveness among other machine learning methods, especially regarding image recognition, because they rely on automatic feature extraction instead of human feature selection [7]. Several studies have proposed methods based on deep learning for the detection and recognition of plant diseases. However, many obstacles prevent the better use of this technology. Indeed, one challenge is the inability to collect dataset images for all diseases across leaf types. Moreover, several types of diseases are fast-spreading, making it challenging to capture them on leaves in time. Additionally, the work [8] showed that the model, during the learning process, is not affected only by the characteristics of the disease but also by the characteristics of the crop. This implies that extracting features from a specific crop and disease cannot be extended to other crops and diseases. Creating a generalized system for plant disease classification requires training the classification model on a dataset containing pairs of all kinds of crops and all kinds of diseases. Unfortunately, such a dataset does not exist, and it is challenging, if not impossible, to build. There are millions of plants and crops in the world, and millions of diseases may affect these plants and crops. This requires time and effort to collect data. These problems directly affect the performance of deep learning-based systems due to the lack of data. The inability to access sufficient datasets obstructs the deep learning system's capacity to generalize patterns learned in plant disease recognition.

To tackle this challenge, this paper introduces a novel deep learning-based system designed to overcome the limitations of specific crop training problems found in the existing literature and generalize the plant disease recognition process to all plant and disease types. Our approach focuses on the core objective of identifying the disease itself, rather than solely relying on the visual appearance of the diseased leaf. It particularly emphasizes recognizing the healthiness or infection in small pieces of the leaf. To accomplish this, we devised a technique where each leaf image in the training dataset is split into smaller patches, effectively isolating the infected areas from the healthy regions. This splitting strategy facilitates the augmentation of healthy data and the extraction of disease-related features from the crops. To ensure efficient and accurate classification and for the outstanding performance demonstrated by the Inception model in recognizing diseases, we employed the small Inception model architecture, renowned for its ability to handle small images while maintaining high performance. Additionally, our proposed system incorporates a feature that allows for the calculation of the disease's prevalence rate on the leaf, providing valuable insights for further analysis and decision-making. This comprehensive approach demonstrates the effectiveness and potential of our system in overcoming the challenges associated with crop-specific training and contributes to the advancement of plant disease recognition.

The rest of the paper is organized as follows: Section 2 presents a summary of the related work. Section 3 defines the proposed methods and the materials used. Section 4 is devoted to the analysis of the obtained results. Section 5 concludes the paper.

2. Related work

In recent years, population growth and decreased resources have prompted researchers to migrate to other techniques for supervising crops and planting, especially for disease detection. Due to their success in other domains, many studies have been carried out on plant and crop disease identification by deep learning. However, detecting plant leaf diseases using deep learning presents several challenges that researchers strive to overcome in the development of efficient and robust models. Firstly, the diversity of plant species and the myriad disease manifestations across different crops pose a significant hurdle in creating a universal model. Deep learning models often struggle to generalize across various plant types and diseases not present in the training dataset. Another challenge is the limited availability of labeled datasets specifically designed for plant leaf diseases, which hampers the training process and can potentially lead to biased models [2]. Furthermore, the dynamic nature of environmental conditions introduces variability in the appearance of healthy and diseased leaves, making it challenging for models to discern subtle differences. Issues related to data imbalance, where certain diseases or crops may be underrepresented, can impact the model's ability to accurately recognize less common diseases. Additionally, the computational resources required for training deep learning models, especially for large-scale architectures, can pose a practical challenge for researchers with limited access to high-performance computing resources [9]. The proposed state-of-the-art methods to deal with this issue can be divided into three categories: identification of a specific disease in a specific crop; identification of several diseases in a specific crop; and the rest cover the identification of several diseases in various crops. Table 1 summarizes the most recent studies in the recognition of plant leaf disease.

2.1. Identification of a specific disease in a specific crop

The works presented in this section focused on the identification of specific diseases in specific crops. In [10], the authors developed a method for detecting Fusarium head blight in wheat. They employed transfer learning with a pre-trained Mask RCNN network and used color image processing to identify disease areas in grayscale images. Their model demonstrated effective detection of diverse disease shapes and sizes. The presented work in [11] proposes a rice blast recognition method based on CNN. The comparative study showed that CNN-based feature extraction outperformed local binary pattern histograms (LBPH) and Haar-WT (Wavelet Transform). Additionally, disease classification using CNN with Softmax and CNN with SVM achieved better results. To think through the problems that appeared in the image's field acquired and the limitations of the data, [12] proposed a new approach that can identify northern leaf blight in corn. They employed a sequence of CNNs to classify image regions and a final CNN to classify the entire image. In [13], the authors presented a model for classifying ginkgo leaf disease using two datasets, one from the field and one from a laboratory. They employed two different CNN architectures, VGG-16 and InceptionV3, and observed varying behavior based on the dataset.

2.2. Identification of several diseases in a specific crop

In this section, we present relevant papers that treat several diseases in only one crop. In [14], the authors developed a tool for tomato leaf disease classification. They used four deep learning networks and fine-tuned them with feature extraction on two datasets: one from external sources and another from laboratory-acquired data. To identify apple leaf diseases, a MobileNet CNN architecture was used in a dataset acquired from agricultural experts in [15]. The work done by [16] presented an effective approach for recognizing four different rice leaf diseases using deep learning and SVM. The method involved image processing, shape and color feature extraction, and SVM-based classification. In [17], they utilized transfer learning with the AlexNet CNN architecture to recognize Camellia oleifera leaf diseases. The study involved experiments with a self-acquired dataset, evaluating both training from scratch and transfer learning from ImageNet. To recognize the leaf blight and the rust in the corn, a VGG16 CNN architecture pre-trained with ImageNet was used in [18]. The authors in [7] presented a novel CNN architecture based on AlexNet and GoogLeNet Inception to identify apple leaf diseases. The results indicated that the proposed CNN has the highest accuracy compared with SVM, BP, AlexNet, GoogLeNet, ResNet20, and VGG16. The authors in [19] suggest a new model, named deconvolution-guided VGG network, for the identification of tomato leaf diseases and the segmentation of disease spots. The proposed model can work with all problems that can appear in images, like low light, shadows, and occlusions. In [20], the authors developed a specialized tomato leaf disease recognition method for Android mobile devices. This approach efficiently distinguishes between various diseases at different stages by adjusting the layers of the AlexNet architecture to create the Multi-Scale AlexNet. To recognize maize leaf diseases from having similar spots in real complicated backgrounds, [21] modified the Faster R-CNN by adding a batch normalization processing layer and introducing a center cost function. The improved model showed its effectiveness in terms of accuracy and detection time compared with other CNN architectures under complex field conditions. In [22], a method for recognizing early and late fine-grained tomato disease was defined. The authors proposed a new convolutional neural network model that uses ARNet architecture combined with attention and residuals for extracting fine region features. A global pooling extended convolutional neural network (GPD CNN) was defined in [23] to identify the cucumber leaf disease. The model is an improvement of the AlexNet architecture to solve the problem of the parameters and the single feature scale. A multi-scale ResNet CNN was proposed in [24] for identifying different degrees of grape diseases with improved accuracy. The model is about the combination of CNN, Mask R-CNN, multi-scale convolutions, ResNet, and SENet. Compared with ResNet, the proposed model scores the highest average accuracy. In [25], the authors introduced a novel CNN architecture for defining and classifying tomato leaf diseases. Their new CNN model outperforms pre-trained VGG16, InceptionV3, and MobileNet in terms of performance. In [26], a new method called INAR-SS was used to early detect apple leaf disease in real-time. The proposed method was developed based on the combination of the SSD with the inception module and rainbow concatenation. The work done in [27] proposes a Plant Pathology Challenge dataset and is used with the ResNet50 CNN network pre-trained on ImageNet to classify four apple leaf diseases. Two experiments were performed: the classification of a single disease and a combination of several diseases. For the sake of detecting the smallest targets of balsam pear leaf diseases in the field conditions, the authors of [28] improved the faster R-CNN by introducing pyramid networks to increase the different sizes of bounding boxes. The authors in [29] proposed a new architecture of ResNet 50 for the detection of three wheat diseases in real acquisition conditions. The data were downsampled into smaller images for early disease detection, and the data was augmented using an artificial background. In [30], a new method for the segmentation of cucumber leaf lesions was proposed. The authors suggest modifying the CNN architecture while avoiding illumination and background. The proposed method proved its performance in the segmentation of disease spots and also reduced the time of training. The authors of [31] proposed a robust deep learning (DL)-based approach. Specifically, they introduced a ResNet-34-based Faster-RCNN for tomato plant leaf disease classification. This approach aims to provide timely localization and identification of various tomato leaf diseases. The aim of the study proposed in [32] is to diagnose tomato leaf diseases by classifying images into healthy and unhealthy categories. This classification task is accomplished using two pre-trained convolutional neural networks (Inception V3 and Inception ResNet V2). In [33] the authors developed an automatic model for classifying and identifying types of diseases based on MobileNet using bean leaf images. The emphasis is on employing an efficient network architecture to build accurate models by evaluating and comparing the effectiveness of different architectures to identify the best one for bean leaf disease classification. The paper [34] presents an end-to-end model for classifying healthy and diseased corn plant leaves, utilizing pre-trained convolutional neural networks EfficientNetB0 and DenseNet121 by considering parameter efficiency through feature fusion and data augmentation techniques. The authors of [35] propose the use of a deep learning architecture based on the EfficientNet convolutional neural network for classifying segmented tomato diseases. The segmentation of images is carried out using a modified U-Net architecture. The authors of [36] are using a combination of deep learning (Inception Net) and semantic segmentation techniques (U-Net and Modified U-Net) to address the task of detecting and segmenting tomato plant diseases. In the study [37], the authors introduced a novel classification network named Dense Inception MobileNet-V2 Parallel Convolutional Block Attention Module Network (DIMPCNET). This network was specifically designed to tackle challenges in identifying tomato leaf diseases, including complexities in backgrounds, subtle differences between diseases, and significant variations within the same diseases. In the work [38], the authors present a method for real-time and non-destructive detection of tomato diseases utilizing an enhanced version of MobileNetV3. The approach involves fine-tuning MobileNetV3 to address the challenge of model overfitting caused by limited data.

2.3. Identification of several diseases in various crops

In a more complex way, the works presented in this category try to detect several diseases in several crops. In [39], the authors introduced a method for identifying plant diseases by focusing on individual lesions and spots. They segmented these lesions and spots from the entire leaf image and conducted experiments for lesion classification and disease detection. This approach enhances data augmentation and enables the detection of multiple diseases on the same leaf. In [40], the authors introduced a system-based disease identification method that focuses on identifying diseases independently of crop type. This approach relies on using common disease names. The system was trained with three methods: trained from scratch, pre-trained with ImageNet, and pre-trained with PlantCLEF2015. The results show that the model can identify diseases in all crops, including crops that have not been seen in the experiment. To identify the plant disease, a novel CNN-based VGGNet pre-trained on Imagenet, the Inception module, and transfer learning was proposed in [41]. The new CNN, INC-VGGN, was compared with DenseNet-201, ResNet-50, Inception V3, and VGGNet-19. In [42], a neural network is proposed to segment the leaf image with high accuracy. The proposed network is decomposed into a regional disease spot detection network and a regional disease spot segmentation network. To segment the disease region accurately, a multi-scale convolution kernel was used. The work presented in [43] allowed for improving the architecture of ResNet by proposing a multi-scale ResNet. The model is used for the identification of vegetable leaf diseases in limited hardware by reducing the size of the model. The multi-scale ResNet has proven its efficiency compared with other models (AlexNet, VGG16, ResNet50, and SqueezeNet). In [44], the authors proposed three different CNN architectures, namely RESNET-MC-1, RESNET-MC-2, and RESNET-MC-3, which integrate non-image data with an image-based convolutional neural network to classify plant diseases in the context of the corresponding crop. This combination allowed for reducing the complexity of the disease classification while maintaining high accuracy. The paper [45] discusses a model architecture for plant disease detection using depthwise separable convolution. The model includes two versions of depthwise separable convolution, specifically Modified MobileNet and Reduced MobileNet. The results obtained from these models are then compared with those from MobileNet. To address the limitations of machine learning models, particularly those related to hardware sophistication, limited scalability, and inefficiency in realistic use. The authors of [46] propose a DeepLens Classification and Detection Model (DCDM) for automatically detecting and classifying leaf diseases in various plants. The proposed DCDM utilizes scalable transfer learning on Amazon Web Services (AWS) SageMaker and is designed for real-time functional usability. In [47] the authors utilized convolutional neural network (CNN)-based pre-trained models for efficient plant disease identification. Specifically, the focus was on fine-tuning the hyperparameters of

Table 1
Recent studies on plant diseases recognition.

Reference	Crops	Diseases	Datasets	Deep Learning models
[10]	Wheat	Fusarium Head Blight	Self	CNN (Mask-RCNN)
[11]	Rice	Rice blast	Self	New CNN
[12]	Corn	Northern leaf blight	Self	CNN (AlexNet, GoogLeNet Inception)
[13]	Ginkgo biloba	Ginkgo leaf	Self	CNN (VGGNet-16, InceptionV3)
[14]	Tomato	6 Diseases	Self	CNN (VGG-16, VGG-19, RestNet, InceptionV3)
[15]	Apple	Alternaria leaf blotch, rust	Self	CNN (MobileNet, RestNet152, InceptionV3)
[16]	Rice	4 diseases	Self	New CNN, SVM
[17]	Camellia oleifera	4 diseases	Self	CNN (AlexNet)
[18]	Corn	Leaf blight, rust	Self	CNN (VGG-16)
[7]	Apple	4 disease	Self	CNN (AlexNet, GoogLeNe Inception)
[19]	Tomato	10 diseases	PlantVillage	CNN (Deconvolution Guided VGGNet)
[20]	Tomato	7 diseases	PlantVillage, Self	CNN (Improved Multi-Scale AlexNet)
[21]	maize	9 diseases	Self	R-CNN (Improved Faster R-CNN)
[22]	Tomato	5 disease	AI challenger	CNN (New CNN ARNet based)
[23]	Cucumber	6 disease	Self	CNN (Global pooling dilated convolutional neural network (GPDCNN))
[24]	Grape	8 disease	AI challenger, Self	CNN (Multi-Scale ResNet)
[25]	Tomato	9 disease	PlantVillage	New CNN
[26]	Apple	5 disease	Self	INAR-SSD
[27]	Apple	4 diseases	Plant Pathology 2020 Challenge, self	CNN (ResNet50)
[28]	Balsam pear	5 disease	Self	R-CNN (Improved Fater R-CNN)
[29]	Wheat	3 diseases	Self	CNN (ResNet50)
[30]	Cucumber	6 diseases	Self	Full convolution neural network
[31]	Tomato	14 diseases	PlantVillage	CNN (ResNet-34-based Faster-RCNN)
[32]	Tomato	3 diseases	PlantVillage, self	CNN (Inception V3, Inception ResNet V2)
[33]	Bean	3 diseases	iBean	CNN (MobileNet, Mobilev2)
[34]	Corn	4 diseases	PlantVillage	CNN (EfficientNetB0, DenseNet121)
[35]	Tomato	10 diseases	PlantVillage	CNN (EfficientNet)
[36]	Tomato	6 diseases	PlantVillage	CNN (Inception Net)
[37]	Tomato	5 diseases	Self	CNN (PCBAM)
[38]	Tomato	10 diseases	PlantVillage	CNN (MobileNetV3)
[39]	14 plant species	79 diseases	Self	CNN (GoogLeNet)
[40]	14 plant species	26 diseases	PlantVillage, IPM, Bing	CNN (GoogLeNet, VGG-16, InceptionV3)
[41]	Rice, corn	9 disease	PlantVillage, self	CNN (INC-VGGN)
[42]	3 crops	9 disease	Self	Regional disease detection network (RD-net)
[43]	4 crops	15 diseases	PlantVillage, AI Challenge2018, self	CNN (Multi-scale ResNet)
[44]	5 crops	17 diseases	Self	CNN (RESNET-MC-1, RESNET-MC-2,RESNETMC-3)
[45]	24 plants	55 diseases	PlantVillage	CNN (Modified MobileNet, Reduced MobileNet)
[46]	6 crops	25 diseases	PlantVillage	CNN (DeepLens)
[47]	14 crops	38 diseases	PlantVillage	CNN (DenseNet-121, ResNet-50, VGG-16, Inception V4)
[48]	14 crops	38 diseases	PlantVillage	CNN (naïve network)
[49]	Bell pepper, potato	2 diseases	PlantVillage, self	CNN (YOLOv5)
[50]	Apple, grape	6 diseases	Crop leaf image data	Vision Transformer (ViT)

popular pre-trained models, including DenseNet-121, ResNet-50, VGG-16, and Inception V4. The paper [48] proposes a methodology for crop disease detection involving labeling infected leaves, pixel-based operations, feature extraction, image segmentation, and disease classification using convolutional neural networks (CNN). The work [49] focused on training a deep model, specifically the YOLOv5 model, to distinguish between healthy and unhealthy leaves. The trained model was subsequently applied to both exclusive and public datasets. The outcomes demonstrate the model's ability to swiftly and accurately identify even small disease patches on plant leaves.

All the above-mentioned works proved that the usage of deep learning in the recognition of plant diseases improved the accuracy. Most of the proposed work tries to address the data deficit by using data augmentation and transfer learning. However, most of the previous

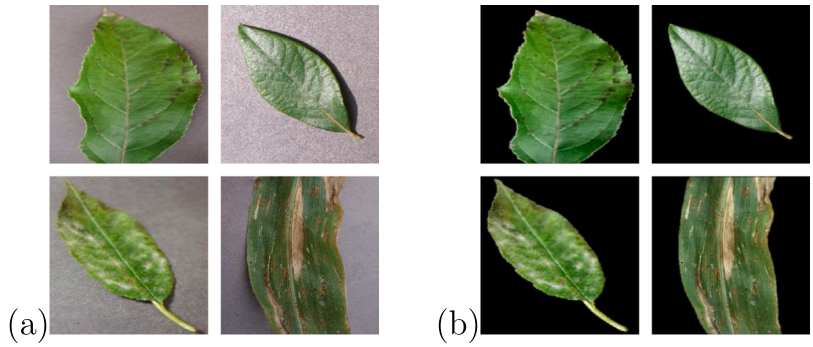

Fig. 1. Examples of PlantVillage dataset images: (a) images with background, (b) images segmented from the background.

work suffers from a lack of robustness and generalization. Indeed, each method specialized in specific crops or diseases contained in the dataset that was used in each paper. Also, the lack of data was not addressed but rather diverted to cover the deficit.

Recently, numerous advanced deep learning methods have been developed, particularly in the field of structural damage detection. This domain shares similarities with disease detection in terms of complexity and variety. Among the most pertinent methods are: Deep learning-based crack damage detection using convolutional neural networks [51], Autonomous structural visual inspection using region-based deep learning for detecting multiple damage types [52], SDDNet: Real-time crack segmentation [53], Efficient attention-based deep encoder and decoder for automatic crack segmentation [54], and Dual encoder–decoder-based deep polyp segmentation network for colonoscopy images [55]. These methods have successfully detected and classified structural damages despite encountering complex issues, highlighting the potential of convolutional neural networks in identifying subtle anomalies. By drawing parallels between challenges in structural damage and leaf disease detection, the methodologies developed for structural damages offer a unique perspective. Indeed, adapting pixel-wise segmentation methods from this domain could prove instrumental in quantifying the size of detected plant diseases, potentially addressing the robustness and generalization challenges encountered in prior works.

3. Materials and methods

3.1. Dataset

PlantVillage dataset [56] is an open-access dataset released through the online platform PlantVillage (www.plantvillage.org) to recognize plant leaf diseases. The PlantVillage dataset was chosen for its diversity, comprehensiveness, annotations, benchmark status, accessibility, and relevance to real-world agricultural challenges and was commonly employed in research. The dataset consists of healthy and diseased leaf images, estimated at 54305 images of 14 crop kinds and 20 disease kinds, divided into 38 classes: 12 healthy leaf classes and 26 leaf diseased classes. The images were acquired in the laboratory with a uniform background and dimensions of 256 × 256 pixels. In our paper, a new version of the PlanVillage dataset created into [57] is used, where the leaves are segmented from the background. Fig. 1 shows examples of images with and without backgrounds.

3.2. The proposed approach

This section provides a comprehensive explanation of the methodology employed to develop a generalized plant disease classification system. Following the training phase, the proposed system can detect the presence of leaf infections without requiring any prior information regarding the specific crop or disease type. Furthermore, the system includes a feature that enables the quantification of the disease's extent within the leaf. The workflow of the proposed system is summarized in Fig. 2, illustrating the key steps involved in its operation.

3.2.1. Generalization of the disease detection process to any crop type

To press the deep learning model during the learning process to not be affected by the crop characteristics and focus solely on extracting disease-related features, we propose a novel method. Instead of detecting disease from the complete leaf, we suggest identifying the disease from small patches representing small pieces of the leaf. We propose to split the leaf images into uniform and equal patches, ensuring the complete elimination of crop type and structure influences. These patches represent small leaf pieces that do not contain any leaf characteristics. This approach ensures that the extracted disease features from a specific crop extend to other crops. Processing small leaf pieces that lack any crop type information enables the generalization of the disease detection process to any crop type, even if it was not seen in the training process.

To apply the proposed method, the image (I) must be split into square patches (P) with dimensions of 32 × 32 pixels. This size was chosen to match the input size of smaller CNN architectures. If the image size is not a multiple of 32, the image is resized to the nearest multiple of 32 to ensure that the system can process any image size, as illustrated in Eqs. (1) and (2).

$$I = \{P_1, P_2, \dots, P_{N\_P}\} \tag{1}$$

$$N\_p = (H/h) \times (W/w) \tag{2}$$

where N_p is the number of patches extracted from one image, H is the width of the image, W is the height of the image, h is the proposed width of the patches, and W is the proposed height of the patches.

Each image in the PlantVillage dataset, originally measuring 256 × 256 pixels, underwent a comprehensive splitting process to create 64 patches, as exemplified in Eq. (4). The splitting was performed by dividing the image into a grid of 8 rows and 8 columns, resulting in each patch having a size of 32 × 32 pixels. This process was implemented by following the procedure outlined in Eq. (3), which provides a mathematical representation of the splitting process.

$$\begin{aligned} N\_p &= (256/32) \times (256/32) \\ N\_p &= 8 \times 8 \\ N\_p &= 64 \end{aligned} \tag{3}$$

$$\begin{aligned} I &= \sum_{i=1}^{8} \sum_{j=1}^{8} P_{ij} \\ I &= \{P_{11}, P_{12}, \dots, P_{88}\} \end{aligned} \tag{4}$$

After image splitting, some criteria were applied to the patches to preserve only the useful ones and eliminate noise in the dataset. After analyzing each patch based on its pixel RGB color, we calculated the patch's black pixel percentage, with RGB values representing black, as presented in Eq. (5).

$$\begin{aligned} B &= \sum_{i=1}^{N} \delta(R_i = 0 \wedge G_i = 0 \wedge B_i = 0) \\ B\_p &= \frac{B}{Total\,Number of\,Pixels} \times 100 \end{aligned} \tag{5}$$

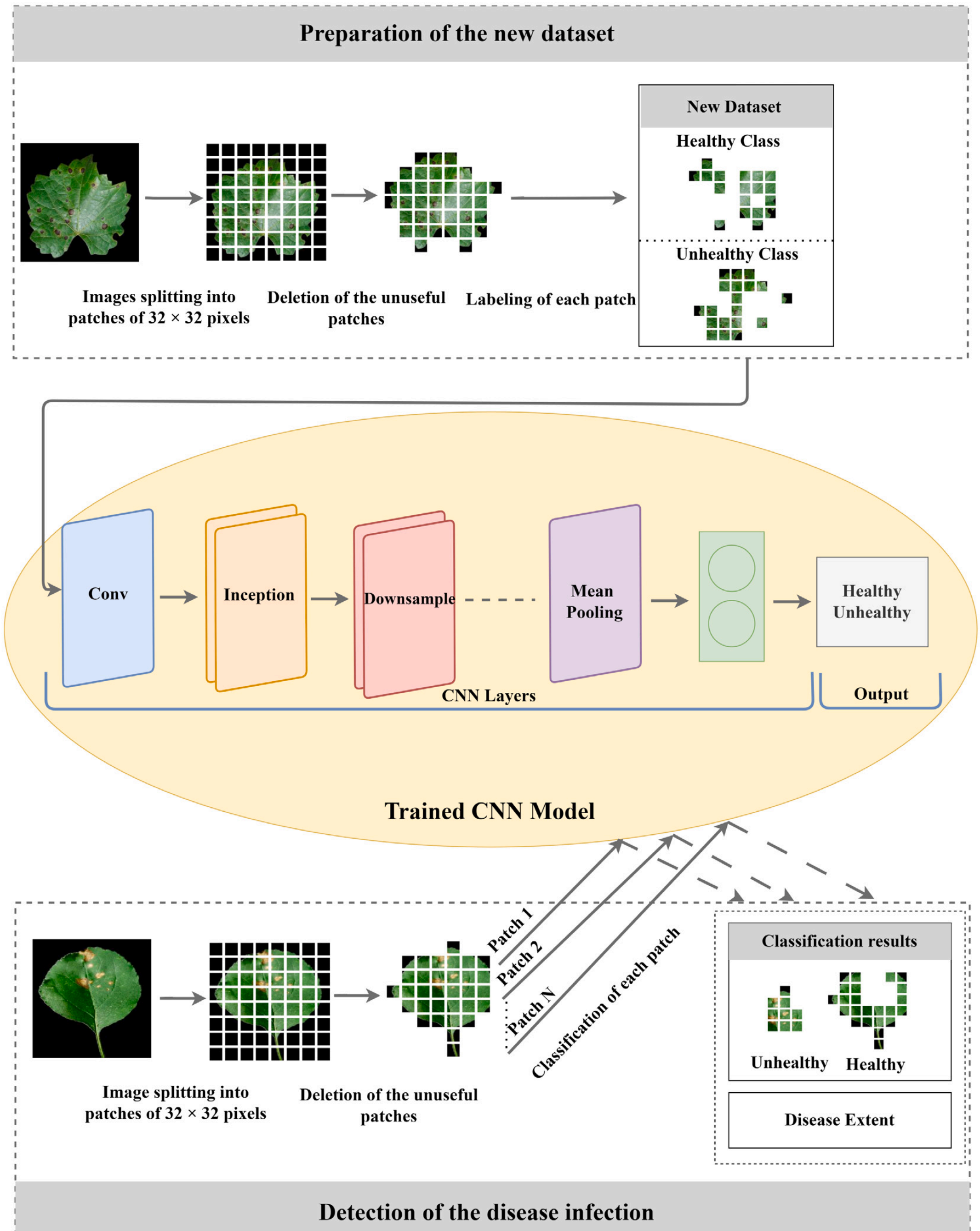

Fig. 2. Flowchart of the proposed model-based convolutional neural network (CNN).

where B is the patch black pixel number, $B_{\_p}$ is the patch black pixel percentage, N is the total number of pixels in the image, $R_{\_i}$, $G_{\_i}$, and $B_{\_i}$ are the red, green, and blue components of the ith pixel, respectively. $\delta()$ is the counter delta function, which is equal to 1 if the condition inside is true and 0 otherwise.

Patches that consisted entirely of black pixels (the black pixels percentage was equal to 100%) signifying leaf pieces' absence, were eliminated. For the remaining patches, to stay as faithful as possible to the PlantVillage dataset, we selectively kept patches with a percentage of black pixels less than or equal to that of the original image, estimated to be approximately 50%. Eq. (6) illustrates the implementation of the process for eliminating unuseful patches. This approach guarantees that the results closely match the characteristics of the initial image, maintaining consistency with the dataset. Fig. 3 illustrates the processes of image splitting and deletion of unuseful patches.

$$I = \{P \mid P \in I, \mathit{and} \text{ black pixels percentage B\_p} \geq \text{selected black pixels percentage}\} \tag{6}$$

3.2.2. Generalization of the disease detection process to any disease type

Due to the insufficient data available for many disease types, it is challenging to detect the presence of these diseases. At the very least, the initial step should focus on detecting the presence of infection without identifying the specific disease type. We propose achieving this by training the system to recognize healthy and unhealthy leaf

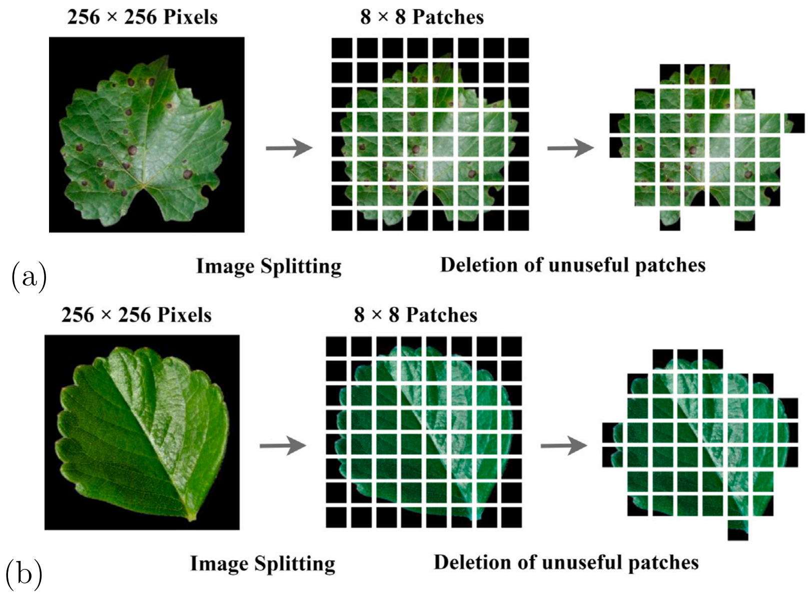

Fig. 3. Image splitting examples: (a) Infected grape leaf image, (b) Healthy strawberry leaf image.

characteristics without considering the disease type. We suggest utilizing healthy patches from all crop types to extract features of healthy leaves and infected patches from all disease types to extract features of unhealthy leaves. This approach enables the distinction between healthy and unhealthy leaves, infected by any disease type, even if not considered in the learning process. Considering all the unhealthy leaf pieces infected with several disease types as one class representing unhealthy leaves results in the generalization of disease detection for any disease type.

This approach was systematically implemented across all split patches from the PlantVillages dataset. The labeling of these patches was conducted through a visual inspection process, which spanned a duration of three months and involved the expertise of professionals in the field. The labeling criteria were as follows: patches presenting healthy leaf pieces, devoid of any disease symptoms, were labeled as 'healthy'. Conversely, patches presenting leaf pieces containing signs of disease were labeled as 'unhealthy', as exemplified in Eq. (7). This labeling strategy allows the healthy set to benefit from the unhealthy leaves because the latter can have healthy parts. Fig. 4 presents the illustration of the labeling process.

$$I = \{P \mid P \in I, and \text{ healthy}\} \cup \{P \mid P \in I, and \text{ infected}\} \tag{7}$$

Applying the proposed methods to the PlantVillage dataset allows for the generation of a novel dataset labeled into two classes: healthy (H) and unhealthy (U), where:

$$H = \sum_{c=1}^{N\_c} \sum_{i=1}^{N\_i\_c} \{P \mid P \in I_i, and \text{ healthy}\} \tag{8}$$

$$U = \sum_{c=1}^{N\_c} \sum_{i=1}^{N\_i\_c} \{P \mid P \in I_i, and \text{ unhealthy}\} \tag{9}$$

where N_c is the number of dataset classes, N_i_c is the number of class C images.

3.2.3. Detecting and determining the extent of the disease

To identify the presence of disease in a leaf, the proposed approach follows the same process outlined in Section 3.2.1. The leaf image is split into non-overlapping patches, each measuring 32 × 32 pixels, and unuseful patches are removed. Subsequently, each retained patch is presented to the classifier to determine whether the corresponding section of the leaf is healthy or diseased (Fig. 2).

This process allows for determining the disease extent on the leaf. The splitting process and the prediction of each patch class enable the assessment of the disease extent on the leaf. The prevalence rate of the disease (P) is calculated by first determining the number of unhealthy patches. Subsequently, the percentage of these patches relative to all the patches that constitute the entire leaf is calculated, as illustrated in Eq. (10).

$$P = \frac{P_U * 100}{P_H + P_U} \tag{10}$$

where P_U is the number of unhealthy patches and P_H is the number of healthy patches.

3.3. The new PlantVillage dataset

The new dataset is composed of 1860316 samples: 846162 healthy and 1014154 unhealthy. The details of the new dataset are illustrated in Table 2. All patches extracted from images of crops infected with the following diseases (huanglongbing (citrus greening), leaf blight (isariopsis leaf spot), powdery mildew) were labeled as unhealthy. This decision was based on the fact that these diseases affect all regions of the leaves. Therefore, all extracted patches from these images were unhealthy.

3.4. CNN network and experimental setup

The CNN architecture employed for classifying the extracted patches from the leaf is the small Inception model. The decision to use the small Inception model was influenced by the success of the GoogLeNet Inception model in the field of plant disease detection. The small Inception architecture is an adapted version of the GoogLeNet Inception architecture specifically designed for small input sizes. It includes a simplified version of the Inception module, as depicted in Fig. 5. The small Inception architecture comprises three modules: the Conv module, the Inception module, and the Downsample module, with details of each module illustrated in the accompanying figure.

Training from scratch was the mechanism used to train the model, implemented in Python with the TensorFlow library. The different hyperparameter values were determined empirically based on the experimental observations of many tests. Indeed, several combinations were explored to select the parameter values that yielded the best performance in the proposed dataset and could maintain the power of

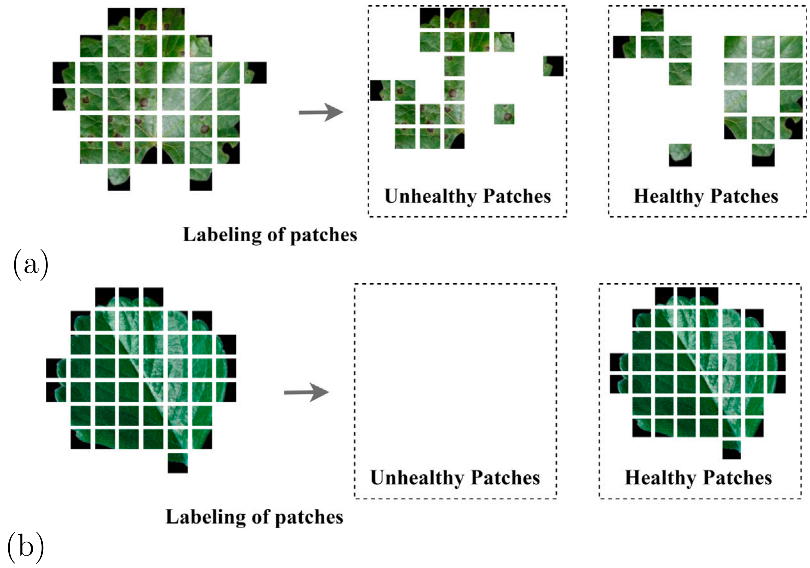

Fig. 4. Data labeling examples: (a) Infected Grape leaf image, (b) Healthy strawberry leaf image.

Table 2
The new PlantVillage dataset details.

Disease name	Crop name	Number of samples	Number of patches	Number of unhealthy patches	Number of healthy patches
Healthy	Apple	1645	48 693	0	48 693
Healthy	Blueberry	1502	38 080	0	38 080
Healthy	Cherry (including sour)	854	28 465	0	28 465
Healthy	Corn (maize)	1162	74 030	0	74 030
Healthy	Grape	423	16 136	0	16 136
Healthy	Peach	360	7081	0	7081
Healthy	Pepper, bell	1478	54 876	0	54 876
Healthy	Potato	152	5505	0	5505
Healthy	Raspberry	371	12 723	0	12 723
Healthy	Soybean	5090	190 448	0	190 448
Healthy	Strawberry	456	16 904	0	16 904
Healthy	Tomato	1591	48 173	0	48 173
Bacterial spot	Peach	2297	60 889	40 187	20 702
Bacterial spot	Pepper, bell	997	36 250	19 097	17 153
Bacterial spot	Tomato	2127	72 774	38 222	34 552
Black rot	Apple	621	20 671	4992	15 679
Black rot	Grape	1180	41 174	18 757	22 417
Cedar apple rust	Apple	275	8104	5831	2273
Cercospora leaf spot gray leaf spot	Corn(maize)	513	25 138	22 738	2400
Common rust	Corn (maize)	1192	59 798	56 887	2911
Early blight	Potato	1000	37 306	19 097	17 153
Early blight	Tomato	1000	30 999	16 850	14 149
Esca(black measles)	Grape	1384	47 236	27 737	19 499
Haunglonbing (Citrus greening)	Orange	5507	194 626	194 626	0
Late bligh	Potato	1000	33 730	16 242	17 488
Late blight	Tomato	1909	49 721	28 769	20 952
Leaf mold	Tomato	952	22 370	14 596	7774
Leaf blight(Isariopsis leaf spot)	Grape	1075	43 106	43 106	0
Leaf scorch	Strawberry	1109	41 062	39 752	1310
North leaf blight	Corn (maize)	985	46 769	40 987	5782
Powdery mildew	Cherry (including sour)	105	35 822	35 822	0
Powdery mildew	Squash	1835	83 457	83 457	0
Septoria leaf spot	Tomato	1771	51 360	34 377	16 983
Spider mites two spotted spider mite	Tomato	1667	41 933	15 110	26 823
Target spot	Tomato	1404	44 800	6086	38 714
Tomato yellow leaf curl virus	Tomato	5357	155 739	155 739	0
Tomato mosaic virus	Tomato	373	7384	5209	2175
Scab	Apple	630	22 984	10 998	11 986
Total	Total	54 305	1 860 316	1 014 154	846 162

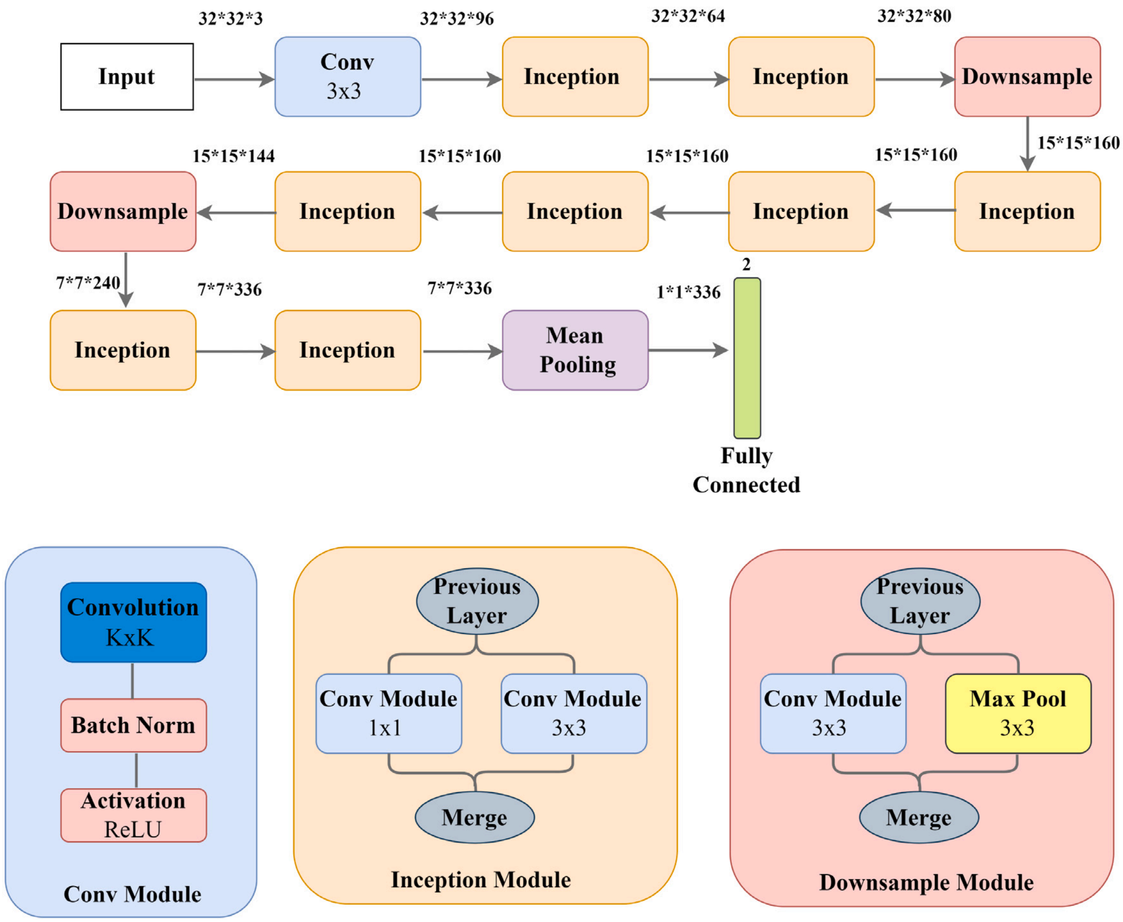

Fig. 5. The small Inception model architecture.

Table 3
Hyperparameters configuration.

Hyperparameter	Values Explored	Selected values
Batch size	16, 32, 64, 128, 256	32
Epochs	1,2, ...,1000	300
Learning rate	0.00005, 0.00002, 0.00001, 0.0005, 0.0002, 0.0001, 0.005, 0.002, 0.001, 0.01	0.001

generalization of the system to other datasets and avoid overfitting. The hyperparameters explored, along with the selected values, are outlined in detail in Table 3. The optimization technique employed was adaptive moment estimation (ADAM) [58], with its default parameter. The experiments were run on Google Colab using the Graphics Processing Unit (GPU).

The dataset was divided into two sets: a training set including 80% of the data from healthy (676929 images) and 80% of the data from unhealthy (811323 images) classes and a testing and validation set including 20% of the data from healthy (169233 images) and 20% of the data from unhealthy (202831 images) classes.

In summary, the proposed methodology employs a deep learning approach, specifically utilizing the small Inception model architecture, to detect and classify plant diseases based on leaf images. The leaf images are systematically split into smaller patches. The extracted patches are labeled as healthy and unhealthy and then used to train the model, enabling the detection of different disease types across different crop types. The prevalence rate of the disease on the leaf is calculated, providing valuable insights into disease distribution. The model is trained from scratch, and various hyperparameter values are explored to optimize performance. The significance of this approach lies in its ability to overcome challenges associated with crop-specific training and contribute to the generalization of plant disease recognition. The following sections will delve into the experimental results and discussions, shedding light on the performance and effectiveness of the proposed system.

4. Results and discussion

In this section, we evaluate the proposed methods for detecting infection regardless of the crop and without specifying the type of disease. Then, we demonstrate the ability of the obtained deep learning model to generalize the infection detection methods to new diseases and crops that were not presented to the classifier in the learning phase. Finally, to properly situate our work in relation to other state-of-the-art methods, we compared the obtained results with the highest-ranked methods on the PlanVillage dataset.

To evaluate the proposed system, we employed various classification performance metrics, including accuracy, loss, and confusion matrix. This choice was imposed since most of the papers that have tested their approaches on the PlantVillage dataset use the same criteria. This simplifies the task of comparison and allows us to properly situate our work in relation to other state-of-the-art methods. Additionally, to tackle imbalanced datasets and conduct a more thorough analysis of the model's performance, we employed various evaluation metrics such as precision, recall, and the F1 score.

Table 4
Performance on training and testing Sets.

	Accuracy	Loss
Train	0.9466	0.1336
Test	0.9404	0.1462

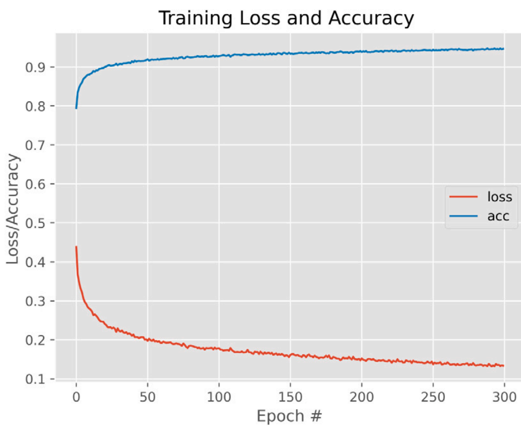

Fig. 6. Training set loss and accuracy progression.

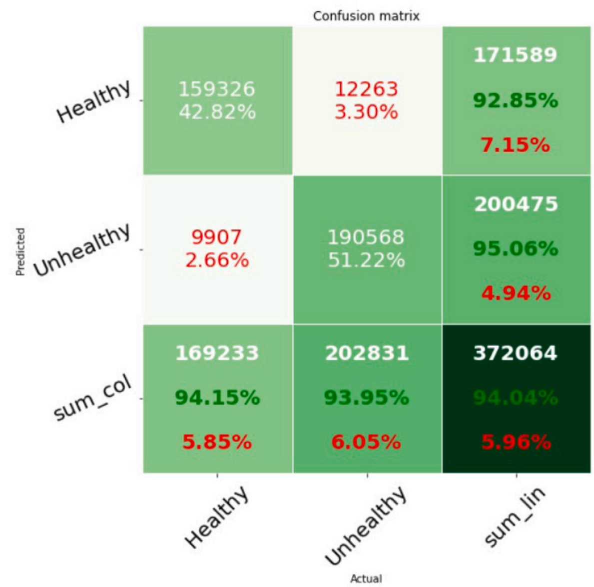

Fig. 7. The New PlantVillage Dataset confusion matrix.

Table 5
Performance on testing set.

	Precision	Recall	F1 score
Test	0.9505	0.9395	0.9450

4.1. Evaluation of the proposed methods

Table 4 presents the accuracy and the loss function obtained for testing and training sets. Fig. 6 defines the progression of the training loss function and accuracy over the training epochs. Fig. 7 is the confusion matrix of the proposed model. From Fig. 6, we can see that the accuracy of the model, in the beginning, progresses quickly, and after a certain number of epochs, the progression starts to stabilize and finally achieves its optimal performance, which is 94,66% after 300 epochs. The loss function takes a reverse curve, where it gradually decreases at the beginning, then becomes stable and, in the end, reaches the minimum value possible.

The testing classification performance of the CNNs was 94.04%. These percentages were distributed among the two classes as follows: the correct prediction from the healthy class was 94.15%; on the other hand, the error was in 5.85% of samples, while in the unhealthy class, the correct predicted samples were 93.95%; and the wrong predicted samples were 6.05%. As we can see, the recognition accuracy of the unhealthy class is better than that of the healthy class. In order to identify the classification failures made by the model, we carefully checked the misclassified patches of each class from the two classes. We noticed that the labeling errors that occurred in the healthy class were due to patches that had been taken from tomato crops. Then, we found that most classification errors that occurred in the unhealthy class were in the classification of patches that had been taken from powdery mildew, huanglongbing (Citrus greening), yellow leaf curl virus, spider mites, two spotted spider mites, and bacterial spot.

The misclassification can be explained as follows: Initially, the similarity between healthy tomato leaves and squash leaves infected with powdery mildew disease is evident, as shown in Fig. 8. This visual resemblance between the appearance of tomato leaves and the symptoms of powdery mildew in squash contributes to patches extracted from tomato crops closely resembling those from the squash powdery mildew class. The confusion results from the fact that the epidermal hairs, or specialized tissues, on tomato leaves can sometimes be mistaken for symptoms of powdery mildew due to their appearance. Both epidermal hairs and powdery mildew can impart a white or powdery look to the leaf surface, creating a potential source of misclassification. Additionally, powdery mildew induces the spread of white powdery spots on the leaves. Patches that included only a few of these white powder spots were not classified as unhealthy, primarily because of the challenge in identifying this limited proportion of white powder spots. Finally, it is noted that some diseases exhibit symptoms characterized by shoots appearing between light green and yellow, as depicted in Fig. 8. These visual symptoms represent a specific set of symptoms associated with various diseases. However, the CNN model faced challenges when processing patches containing smaller shoots, leading to confusion in its classification.

The new dataset shows some imbalance. To investigate the impact of this imbalance and conduct a more thorough analysis of the model's performance, especially in distinguishing between diseased and healthy plants, we evaluated the model using three key metrics: precision, recall, and F1 score. Table 5 displays the results for these metrics. The close alignment of precision, recall, and F1 score indicates a balanced performance of the classification model. A recall of 0.9395 indicates the model's ability to correctly identify around 93.95% of actual positive cases. With a precision of 0.9505, the model demonstrates accuracy by correctly predicting the positive class approximately 95.05% of the time. An F1 score of 0.9450 indicates a model that performs well, with a robust balance between precision and recall. Importantly, the close proximity of these metrics suggests that the model's performance remains consistent and robust to the imbalances in the dataset.

4.2. Evaluation on new crop and new disease

To demonstrate the generalizability of the proposed model, it should be tested on other unknown datasets. The model needs to be evaluated on new crops and diseases that do not exist in the PlantVillage dataset. For this purpose, we selected a crop infected with a disease that was not seen in the training process, such as cotton infected with Soreshin disease. Additionally, we selected a crop that was seen in the training process but infected with an unseen disease, such as soybean infected with southern blight disease. Lastly, we selected an unseen crop with a seen disease, such as coffee infected with rust disease. From the challenging dataset Plant Disease Symptoms (PDDB) [59] we randomly

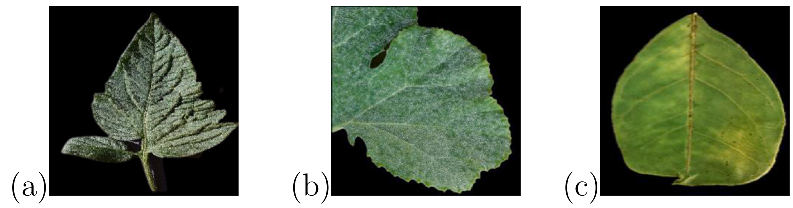

Fig. 8. Data Examples: (a) Healthy tomato leaf image, (b) Powdery mildew-infected squash leaf image, (c) Huanglongbing (Citrus greening) infected orange leaf image.

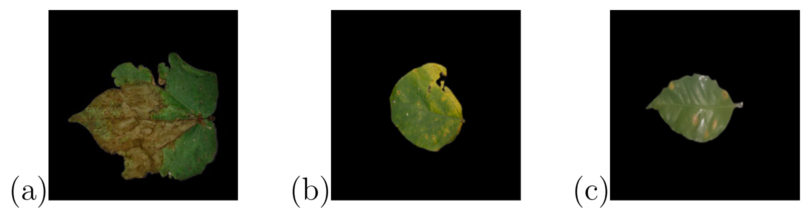

Fig. 9. PDDB dataset examples: (a) Cotton leaf (b) Soybean leaf (c) Coffee leaf.

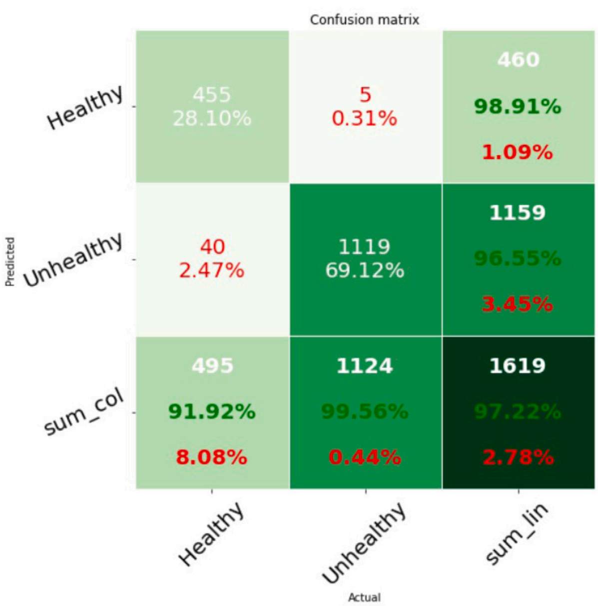

Fig. 10. PDDB dataset confusion matrix.

selected 108 images: 19 cotton images, 62 soybean images, and 27 coffee images. These images underwent the same pre-processing steps as the PlantVillage dataset. After splitting, we obtained 1619 patches, including 495 healthy patches and 1124 unhealthy patches. Fig. 9 gives examples of PDDB dataset images.

As illustrated in Fig. 10, the results showed that the model achieved an accuracy of 97.22%. The recognition accuracies of each class are 91.91% for the healthy class and 99.55% for the unhealthy class. As we can see, the recognition accuracy of the unhealthy class is better than that of the healthy class. To analyze the classification failures made by the system, we carefully examined the misclassified images of the PDDB dataset patches. We noticed that the labeling errors were primarily due to confusion about patches that contained only small-sized disease-symptom spots. For the healthy patches, the confusion was in the light green patches because of their similarity to the shoots appearing between light green and yellow caused by some diseases. These factors were the main causes of misclassification compared to the well-classified images.

4.3. Comparison with other state-of-the-art methods

The comparison of the obtained results with the highest-ranked state-of-the-art methods on the PlantVillage dataset is presented in Table 6. While the proposed method may not have achieved the top accuracy score, it does demonstrate robust and strong performance, showcasing its effectiveness in the context of the evaluated methods. Unlike other state-of-the-art methods that were limited to specific portions of the dataset, our method covers the entirety of the PlantVillage dataset, providing comprehensive coverage and enhancing its applicability. However, what truly distinguishes this system is its exceptional ability to identify new types of crops and diseases that were not included in the training dataset. The proposed method is capable of recognizing infections in any type of crop leaf, regardless of the disease present. This makes it a versatile and multi-crop, multi-disease solution. In contrast, other state-of-the-art methods lack the capability to handle new crops not seen in the training process, limiting their effectiveness as truly multi-crop and multi-disease solutions. The proposed method, by contrast, showcases a higher level of adaptability and robustness in addressing a wide range of crops and diseases.

4.4. Discussion

Based on the findings mentioned above, several deductions can be made. Firstly, the majority of errors stem from the resemblance between specific disease symptoms and distinct healthy crop characteristics. This is especially evident in the challenge of differentiating the light green color present in healthy patches from the shoots that appear between light green and yellow, caused by certain diseases. Additionally, errors are observed in the detection of small disease spots within the patches.

Secondly, the proposed method demonstrated remarkable generalization results. The model successfully applied the extracted features from the training dataset to identify new types of crops and diseases not present in the initial training dataset. This indicates the model's robustness and capacity to adapt to diverse agricultural scenarios beyond its training scope.

Finally, we can conclude that the proposed system performs well despite the diversity of the dataset. Indeed, each class of the dataset consists of images of patches of leaves with different shapes and textures. In addition, the pattern of veins in the leaf varies significantly from one image to another. When it comes to unhealthy samples, there are many diseases, and each disease has its symptoms. Fig. 11 presents a set of examples of the samples within the dataset. Having a general system that can detect the presence of infection in crops

Table 6
A comparative study with state-of-the-art methods using the PlantVillage dataset.

Reference	All dataset	Multi-crop	Multi-disease	Generalized to a new crop	Generalized to a new disease	Accuracy (mean)
[43]	√	×	×	×	×	95.95%
[19]	×	×	×	×	×	93.25%
[20]	×	×	×	×	×	92.7%
[25]	×	×	×	×	×	91.2%
[40]	√	√	×	√	×	91.97%
[41]	×	×	×	×	×	92%
[45]	√	×	×	×	×	98.34%
[31]	×	×	×	×	×	99.97%
[46]	×	×	×	×	×	98.78%
[47]	√	×	×	×	×	99.81%
[32]	×	×	×	×	×	99.22%
[34]	×	×	×	×	×	98.56%
[35]	×	×	×	×	×	99.89%
[36]	×	×	×	×	×	99.12%
[48]	√	×	×	×	×	90.40%
[49]	×	×	×	×	×	93%
[38]	×	×	×	×	×	98.25%
Proposed	√	√	√	√	√	94,04%

Fig. 11. Dataset samples: Examples.

without being dependent on the specific type of crop or disease is crucial. Such a system provides a versatile and widely applicable solution for agriculture. It allows for broader use across various crops and diseases, making it more accessible and practical for farmers and agricultural practitioners. This general system is particularly beneficial in real-world farming scenarios where different crops are cultivated, and various diseases may pose a threat. It contributes to early detection and effective management of crop infections across different contexts.

In conclusion, this section has delved into key findings and their implications. The majority of errors observed in the system stem from challenges in distinguishing disease symptoms from healthy crop characteristics, particularly in cases where there is a resemblance between the light green color in healthy patches and spots appearing between light green and yellow caused by certain diseases. Additionally, challenges arise in accurately detecting small disease spots within patches. The upcoming section will delve deeper into the conclusions derived from the proposed system, providing a more detailed discussion of future work and its implications.

5. Conclusion

This paper addresses the plant disease recognition problem with a new approach based on a smaller input version of a deep convolutional network called the small Inception. The proposed system is able to distinguish between diseased and healthy leaves, regardless of the kind of leaf and disease. This is due to an extra step added before classification that allows processing patches of the leaf instead of the whole leaf. Indeed, the image splitting allowed not only to augment the dataset with real data but also to free up the classifier from any dependence on the leaf structure. In addition, the system can generalize the detection of the infection that is learned on a very specific type of crop and diseases on all the leaves and disease types, even if the latter were not presented to the classifier in the learning phase.

Furthermore, the system showcases an impressive capacity for generalization, enabling the detection of infections across various leaf and disease types, even those not encountered during the learning phase. Training and testing on the PlantVillage dataset and validation on the PDDB dataset underscore the system's robustness and its ability to generalize to previously unseen data. Notably, the proposed system outperforms other state-of-the-art methods, utilizing a single model for all leaf and disease types, a distinctive characteristic compared to specialized methods tailored for specific diseases or crops.

In conclusion, this study introduces an innovative approach to identifying plant diseases, showcasing superior performance and remarkable generalization capabilities. The findings hold significant implications for advancing automated plant disease detection and management, offering an efficient solution for various agricultural scenarios. However, it is crucial to acknowledge potential challenges that may impact the system's performance. The system's adaptability in adapting to diverse environmental conditions could impact the system's robustness in real-world scenarios. As we move forward, future research should explore optimization strategies to enhance the model's performance and adaptability in real-world scenarios. The use of deep learning crack damage detection systems can also be investigated as a potential solution to generalize disease detection in any crop type. Additionally, the adoption of a sliding window approach, as an alternative to our current segmentation method, holds the potential to generate multiple segments, consequently enhancing the system's capacity for disease detection

CRediT authorship contribution statement

Imane Bouacida: Conceptualization, Data curation, Software, Writing – original draft. **Brahim Farou:** Conceptualization, Methodology, Resources, Supervision, Review & editing. **Lynda Djakhdjakha:** Conceptualization, Supervision, Review & editing. **Hamid Seridi:** Conceptualization, Formal analysis, Funding acquisition, Project administration, Validation. **Muhammet Kurulay:** Conceptualization, Investigation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The research data is not available for sharing.

References

[1] Dutta K, Talukdar D, Bora SS. Segmentation of unhealthy leaves in cruciferous crops for early disease detection using vegetative indices and Otsu thresholding of aerial images. Meas: J Int Meas Confed 2022;189:110478. http://dx.doi.org/10.1016/j.measurement.2021.110478.

[2] Li L, Zhang S, Wang B. Plant disease detection and classification by deep learning—A review. IEEE Access 2021;9:56683–98. http://dx.doi.org/10.1109/ACCESS.2021.3069646.

[3] Ramcharan A, Baranowski K, McCloskey P, Ahmed B, Legg J, Hughes DP. Deep learning for image-based Cassava disease detection. Front Plant Sci 2017;8:1852. http://dx.doi.org/10.3389/fpls.2017.01852.

[4] Attri I, Awasthi LK, Sharma TP, Rathee P. A review of deep learning techniques used in agriculture. Ecol Inform 2023;102217. http://dx.doi.org/10.1016/j.ecoinf.2023.102217.

[5] Attri I, Awasthi LK, Sharma TP. Machine learning in agriculture: A review of crop management applications. Multimed Tools Appl 2023;1–41. http://dx.doi.org/10.1007/s11042-023-16105-2.

[6] Shoaib M, Shah B, Ei-Sappagh S, Ali A, Ullah A, Alenezi F, et al. An advanced deep learning models-based plant disease detection: A review of recent research. Front Plant Sci 2023;14:1158933. http://dx.doi.org/10.3389/fpls.2023.1158933.

[7] Liu B, Zhang Y, He D, Li Y. Identification of Apple leaf diseases based on deep convolutional neural networks. Symmetry 2018;10(1):11. http://dx.doi.org/10.3390/sym10010011.

[8] Yosuke T, Fumio O. How convolutional neural networks diagnose plant disease. Plant Phenom 2019;2019. http://dx.doi.org/10.34133/2019/9237136.

[9] Liu J, Wang X. Plant diseases and pests detection based on deep learning: A review. Plant Methods 2021;17:1–18. http://dx.doi.org/10.1186/s13007-021-00722-9.

[10] Qiu R, Yang C, Moghimi A, Zhang M, Steffenson BJ, Hirsch CD. Detection of Fusarium head blight in wheat using a deep neural network and color imaging. Remote Sens 2019;11(22). http://dx.doi.org/10.3390/rs11222658.

[11] Liang W-j, Zhang H, Zhang G-f, Cao H-x. Rice blast disease recognition using a deep convolutional neural network. Sci Rep 2019;9(2869). http://dx.doi.org/10.1038/s41598-019-38966-0.

[12] DeChant C, Wiesner-Hanks T, Chen S, Stewart EL, Yosinski J, Gore MA, et al. Automated identification of northern leaf blight-infected maize plants from field imagery using deep learning. Phytopathology 2017;107(11):1426–32. http://dx.doi.org/10.1094/phyto-11-16-0417-r.

[13] Li K, Lin J, Liu J, Zhao Y. Using deep learning for image-based different degrees of Ginkgo leaf disease classification. Information 2020;11(2):95. http://dx.doi.org/10.3390/info11020095.

[14] Ahmad I, Hamid M, Yousaf S, Shah ST, Ahmad MO. Optimizing pretrained convolutional neural networks for Tomato leaf disease detection. Complexity 2020;2020:1–6. http://dx.doi.org/10.1155/2020/8812019.

[15] Bi C, Wang J, Duan Y, Fu B, Kang J-R, Shi Y. Mobilenet based Apple leaf diseases identification. Mob Netw Appl 2020;27:172–80. http://dx.doi.org/10.1007/s11036-020-01640-1.

[16] Jiang F, Lu Y, Chen Y, Cai D, Li G. Image recognition of four rice leaf diseases based on deep learning and support vector machine. Comput Electron Agric 2020;179:105824. http://dx.doi.org/10.1016/j.compag.2020.105824.

[17] Long M, Ouyang C, Liu H, Fu Q. Image recognition of Camellia Oleifera diseases based on convolutional neural network & transfer learning. Nongye Gongcheng Xuebao/Trans Chin Soc Agric Eng 2018;34:194–201. http://dx.doi.org/10.11975/j.issn.1002-6819.2018.18.024.

[18] Xu J, Shao M, Wang Y, Han W. Recognition of Corn leaf spot and rust based on transfer learning with convolutional neural network. Trans Chin Soc Agric Mach 2020;51(2):230–6. http://dx.doi.org/10.6041/j.issn.1000-1298.2020.02.025.

[19] Ren S, Jia F, Gu X, Yuan P, Xue W, Xu H. Recognition and segmentation model of Tomato leaf diseases based on deconvolution-guiding. Trans Chin Soc Agric Eng 2020;36(12):186–95. http://dx.doi.org/10.11975/j.issn.1002-6819.2020.12.023.

[20] Guo X, Fan T, Shu X. Tomato leaf diseases recognition based on improved multi-scale AlexNet. Trans Chin Soc Agric Eng 2019;35(13):162–9. http://dx.doi.org/10.11975/j.issn.1002-6819.2019.13.018.

[21] Fan X, Zhou J, Xu Y. Recognition of field Maize leaf diseases based on improved regional convolutional neural network. J South Chin Agricult Univ 2020;41(6):82–91. http://dx.doi.org/10.7671/j.issn.1001-411X.202008022.

[22] Hu Z, Yang H, Huang J, Xie Q. Fine-grained Tomato disease recognition based on attention residual mechanism. J South Chin Agricult Univ 2019;40(6):124–32. http://dx.doi.org/10.7671/j.issn.1001-411X.201812048.

[23] Zhang S, Zhang S, Zhang C, Wang X, Shi Y. Cucumber leaf disease identification with global pooling dilated convolutional neural network. Comput Electron Agric 2019;162:422–30. http://dx.doi.org/10.1016/j.compag.2019.03.012.

[24] Xin H, ShuQin L, Bin L. Identification of Grape leaf diseases based on multi-scale residual neural network. Comput Eng 2021;47(5):285–91.

[25] Agarwal M, Singh A, Arjaria SK, Sinha A, Gupta SK. ToLeD: Tomato leaf disease detection using convolution neural network. Procedia Comput Sci 2020;167:293–301. http://dx.doi.org/10.1016/j.procs.2020.03.225, International Conference on Computational Intelligence and Data Science.

[26] Jiang P, Chen Y, Liu B, He D, Liang C. Real-time detection of Apple leaf diseases using deep learning approach based on improved convolutional neural networks. IEEE Access 2019;7:59069–80. http://dx.doi.org/10.1109/ACCESS.2019.2914929.

[27] Thapa R, Snavely N, Belongie S, Khan A. The plant pathology 2020 challenge dataset to classify Foliar disease of Apples. Appl Plant Sci 2020;3(11958). http://dx.doi.org/10.48550/arXiv.2004.11958.

[28] Li J, Lin L, Tian K, Alaa AA. Detection of leaf diseases of Balsam pear in the field based on improved faster R-CNN. Trans Chin Soc Agricult Eng 2020;36(12):179–85. http://dx.doi.org/10.11975/j.issn.1002-6819.2020.12.022.

[29] Picon A, Alvarez-Gila A, Seitz M, Ortiz-Barredo A, Echazarra J, Johannes A. Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild. Comput Electron Agric 2019;161:280–90. http://dx.doi.org/10.1016/j.compag.2018.04.002.

[30] Zhen W, Shanwen Z, Xianfeng W. Method for segmentation of Cucumber leaf lesions based on improved full convolution neural network. Jiangsu J Agricult Sci 2019;35(5):1054–60.

[31] Nawaz M, Nazir T, Javed A, Masood M, Rashid J, Kim J, et al. A robust deep learning approach for Tomato plant leaf disease localization and classification. Sci Rep 2022;12(1):18568. http://dx.doi.org/10.1038/s41598-022-21498-5.

[32] Saeed A, Abdel-Aziz A, Mossad A, Abdelhamid MA, Alkhaled AY, Mayhoub M. Smart detection of Tomato leaf diseases using transfer learning-based convolutional neural networks. Agric 2023;13(1):139. http://dx.doi.org/10.3390/agriculture13010139.

[33] Elfatimi E, Eryigit R, Elfatimi L. Beans leaf diseases classification using MobileNet models. IEEE Access 2022;10:9471–82. http://dx.doi.org/10.1109/ACCESS.2022.3142817.

[34] Amin H, Darwish A, Hassanien AE, Soliman M. End-to-end deep learning model for Corn leaf disease classification. IEEE Access 2022;10:31103–15. http://dx.doi.org/10.1109/ACCESS.2022.3159678.

[35] Chowdhury ME, Rahman T, Khandakar A, Ayari MA, Khan AU, Khan MS, et al. Automatic and reliable leaf disease detection using deep learning techniques. J Agric Eng 2021;3(2):294–312. http://dx.doi.org/10.3390/agriengineering3020020.

[36] Shoaib M, Hussain T, Shah B, Ullah I, Shah SM, Ali F, et al. Deep learning-based segmentation and classification of leaf images for detection of Tomato plant disease. Front Plant Sci 2022;13:1031748. http://dx.doi.org/10.3389/fpls.2022.1031748.

[37] Peng D, Li W, Zhao H, Zhou G, Cai C. Recognition of Tomato leaf diseases based on DIMPCNET. Agron J 2023;13(7):1812. http://dx.doi.org/10.3390/agronomy13071812.

[38] Qiaoli Z, Li M, Liying C, Helong Y. Identification of Tomato leaf diseases based on improved lightweight convolutional neural networks MobileNetV3. Smart Agric 2022;4(1):47. http://dx.doi.org/10.12133/j.smartag.SA202202003.

[39] Barbedo JGA. Plant disease identification from individual lesions and spots using deep learning. Biosyst Eng 2019;180:96–107. http://dx.doi.org/10.1016/j.biosystemseng.2019.02.002.

[40] Lee SH, Goëau H, Bonnet P, Joly A. New perspectives on plant disease characterization based on deep learning. Comput Electron Agric 2020;170:105220. http://dx.doi.org/10.1016/j.compag.2020.105220.

[41] Chen J, Chen J, Zhang D, Sun Y, Nanehkaran Y. Using deep transfer learning for image-based plant disease identification. Comput Electron Agric 2020;173:105393. http://dx.doi.org/10.1016/j.compag.2020.105393.

[42] Zhen W, Shanwen Z, Baoping Z. Crop diseases leaf segmentation method based on Cascade convolutional neural network. Comput Eng Appl 2020;56(15):242–50. http://dx.doi.org/10.3778/j.issn.1002-8331.1905-0193.

[43] Wang C, Zhou J, Wu H, Teng G, Zhao C, Li J. Identification of vegetable leaf diseases based on improved multi-scale ResNet. Trans Chin Soc Agric Eng 2020;36(20):209–17. http://dx.doi.org/10.11975/j.issn.1002-6819.2020.20.025.

[44] Picon A, Seitz M, Alvarez-Gila A, Mohnke P, Ortiz-Barredo A, Echazarra J. Crop conditional convolutional neural networks for massive multi-crop plant disease classification over cell phone acquired images taken on real field conditions. Comput Electron Agric 2019;167:105093. http://dx.doi.org/10.1016/j.compag.2019.105093.

[45] Kamal K, Yin Z, Wu M, Wu Z. Depthwise separable convolution architectures for plant disease classification. Comput Electron Agric 2019;165:104948. http://dx.doi.org/10.1016/j.compag.2019.104948.

[46] Khan A, Nawaz U, Ulhaq A, Robinson RW. Real-time plant health assessment via implementing cloud-based scalable transfer learning on AWS DeepLens. Plos one 2020;15(12):e0243243. http://dx.doi.org/10.1371/journal.pone.0243243.

[47] Eunice J, Popescu DE, Chowdary MK, Hemanth J. Deep learning-based leaf disease detection in crops using images for agricultural applications. Agron J 2022;12(10):2395. http://dx.doi.org/10.3390/agronomy12102395.

[48] Panchal AV, Patel SC, Bagyalakshmi K, Kumar P, Khan IR, Soni M. Image-based plant diseases detection using deep learning. Mater Today: Proc 2023;80:3500–6. http://dx.doi.org/10.1016/j.matpr.2021.07.281.

[49] Khalid M, Sarfraz MS, Iqbal U, Aftab MU, Niedbała G, Rauf HT. Real-time plant health detection using deep convolutional neural networks. Agric 2023;13(2):510. http://dx.doi.org/10.3390/agriculture13020510.

[50] Fu X, Ma Q, Yang F, Zhang C, Zhao X, Chang F, et al. Crop pest image recognition based on the improved ViT method. Inf Process Agric 2023. http://dx.doi.org/10.1016/j.inpa.2023.02.007.

[51] Cha Y-J, Choi W, Büyüköztürk O. Deep learning-based crack damage detection using convolutional neural networks. Comput-Aided Civ Infrastruct Eng 2017;32(5):361–78. http://dx.doi.org/10.1111/mice.12263.

[52] Cha Y-J, Choi W, Suh G, Mahmoudkhani S, Büyüköztürk O. Autonomous structural visual inspection using region-based deep learning for detecting multiple damage types. Comput-Aided Civ Infrastruct Eng 2018;33(9):731–47. http://dx.doi.org/10.1111/mice.12334.

[53] Choi W, Cha Y-J. SDDNet: Real-time crack segmentation. IEEE Trans Ind Electron 2019;67(9):8016–25. http://dx.doi.org/10.1109/TIE.2019.2945265.

[54] Kang DH, Cha Y-J. Efficient attention-based deep encoder and decoder for automatic crack segmentation. Struct Health Monit 2022;21(5):2190–205. http://dx.doi.org/10.1177/1475921721105377.

[55] Lewis J, Cha Y-J, Kim J. Dual encoder–decoder-based deep polyp segmentation network for colonoscopy images. Sci Rep 2023;13(1):1183. http://dx.doi.org/10.1038/s41598-023-28530-2.

[56] Hughes DP, Salathe M. An open access repository of images on plant health to enable the development of mobile disease diagnostics. 2015, http://dx.doi.org/10.48550/ARXIV.1511.08060, arXiv.

[57] Mohanty SP, Hughes DP, Salathé M. Using deep learning for image-based plant disease detection. Front Plant Sci 2016;7:1419. http://dx.doi.org/10.3389/fpls.2016.01419.

[58] Kingma DP, Ba J. Adam: A method for stochastic optimization. 2014, http://dx.doi.org/10.48550/arXiv.1412.6980, arXiv preprint arXiv:1412.6980.

[59] Barbedo JGA, Koenigkan LV, Halfeld-Vieira BA, Costa RV, Nechet KL, Godoy CV, et al. Annotated plant pathology databases for image-based detection and recognition of diseases. IEEE Lat Am Trans 2018;16(6):1749–57. http://dx.doi.org/10.1109/TLA.2018.8444395.